

ISHAN SHAH

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PROFILE

Where privacy protections fail in practice is the central question driving my research. I study **Web Privacy, Privacy Compliance, and Security** through large-scale empirical measurement, examining the gap between what digital services disclose, what regulations require, and what actually happens in practice. My research aims to uncover systematic and overlooked privacy risks, particularly where digital services evolve faster than the mechanisms designed to protect users.

PUBLICATIONS

- **Cookies Are Not the Whole Story: Discrepancies Between Tracking Disclosures and Tracking Behavior.**
Ishan Shah, Ali Rasaii, Anja Feldmann, Devashish Gosain, Ha Dao.
Accepted for publication in the proceedings of the *ACM International Conference on emerging Networking EXperiments and Technologies (CoNEXT 2026)*.
Paper: [Link](#)
- **Two Stories, One Website: Measuring Inconsistencies Between Cookie Banners and Privacy Policies.**
Palak Chopra*, Ishan Shah*, Devashish Gosain, Ha Dao.
Under review at the *USENIX Security Symposium 2027*.
*(Joint First-Authors)

EXPERIENCE

Research Scholar @Max Planck Institute for Informatics (MPII)

1 June 2025 - Present

Supervisor: Prof. Anja Feldmann, MPII, Germany

14 Months

- Investigated gaps between tracking disclosures and actual tracking, going beyond cookies to examine whether consent interfaces accurately represent the broader tracking techniques deployed in practice, including advanced browser-fingerprinting methods.
- Developed a zero-shot prompting approach leveraging contextual understanding of banner text to identify tracking disclosures with 96.8% accuracy across the Tranco Top 100K websites.
- Measured post-rejection fingerprinting activity across 7 geographic vantage points, identifying geographic patterns in continued tracking despite consent rejection and cookie blocking.
- Extended the OpenWPM measurement framework to close interception gaps across browser-fingerprinting techniques, improving coverage by 30%; findings led to a **first-author paper accepted at ACM CoNEXT 2026**.

Research Intern @Trust Lab

Dec. 2024 - July 2025

Supervisor: Prof. Devashish Gosain, IIT Bombay

8 Months

- Uncovered a previously overlooked gap in privacy compliance auditing: consent interfaces can contradict privacy policies, potentially leading users to make privacy choices based on inaccurate or incomplete information about data processing.
- Built BannerPriv, a web-scale framework jointly collecting multi-layer consent interfaces and privacy policies, achieving 94% precision, 88% recall in contradiction detection.
- Developed a 3-stage few-shot LLM pipeline to extract and align disclosures from consent interfaces and privacy policies, identifying 7 potential categories of inconsistencies and highlighting the need for a cross-disclosure consistency check in privacy compliance assessments.
- Examined Tranco Top-50K websites across CMPs, rankings, and website categories, characterizing previously overlooked consent-policy inconsistencies at scale; **first-author paper under review at USENIX Security 2027**.

EDUCATION

Indian Institute of Technology, Jodhpur

2023 - Present

Bachelor of Technology in Artificial Intelligence and Data Science / 4-Year Program

India

Joint Entrance Examination (JEE)

2023

JEE Main: 96.7 Percentile, JEE Advanced: 95.4 Percentile

India

St. Joseph School (CBSE)

2023

Class XII – Percentage: 88.4%

India

St. Joseph School (CBSE)

2022

Class XI – Percentage: 92.4%

India

PROJECTS

Evaluating the Risk of Honeypots Becoming Part of the Attack | [GitHub](#)

Feb. 2025

- Designed and deployed 4 vulnerable honeypots in Docker, enabling covert attacker persistence, security evasion, and 100% untraceable operations.
- Developed HTTP/HTTPS, SSH, FTP attack payloads using Flask & Socket Programming.
- Simulated attacks across 5 protocols (SMB/DNS included) at 1,000+ RPS to test system resilience.
- Exposed telemetry gaps in Cowrie/Dionaea through custom evasion techniques, creating detection blindspots.

OAuth2 Misuse & Bypass Testing Toolkit | [GitHub](#)

Mar. 2025

- Replicated & validated OAuth2 security flaws (including CWE - 601) in controlled environment, weaponizing prior vulnerability reports.
- Tested & confirmed 20 bypass techniques against domain whitelisting/blacklisting defenses, documenting security gaps with MITRE T1071 mappings.
- Compiled detailed technical report covering attack vectors and 2 mitigation strategies, including strict redirect URI and PKCE (RFC 7636).

TECHNICAL SKILLS

Privacy Regulation: GDPR, CCPA

Privacy & Security Frameworks: IAB TCF, ePrivacy, NIST CSF

AI & Machine Learning: PyTorch, scikit-learn, NLP, LLMs, Computer Vision, AI Agents

Web Measurement & Automation: OpenWPM, Selenium, Playwright, BeautifulSoup

Programming: Python, C, C++, JavaScript, Bash, SQL

Systems & Infrastructure: Linux, Windows, Docker, Google Cloud Platform (GCP), VirtualBox

Security: Burp Suite, Metasploit, Nmap, Wireshark, Nessus, Snort, Splunk, Nikto, Dirbuster, Recon-ng, Maltego, SpiderFoot, theHarvester

CERTIFICATIONS

Junior Web Pentester: TryHackMe

Security and Privacy in the Tor Network: University of Colorado System

Cybersecurity Professional Certificate: Google

NIST Cybersecurity Framework (CSF): Infosec

COURSEWORK

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|---|-------------------------------------|
| • Data Structures & Algorithms | • Optimization for Machine Learning |
| • Pattern Recognition & Machine Learning | • Design and Analysis of Algorithms |
| • Computer Systems | • Data Engineering |
| • Computer Networks | • Signals & Systems |
| • Probability Statistics & Stochastic process | • Deep Learning |
| • Maths for Computing | • Dependable AI |
| • Artificial Intelligence | • Computer Vision |

ACHIEVEMENTS & LEADERSHIP

- Ranked **top 6% globally** on TryHackMe; **AIR 3615** among Indian participants.
- Awarded the *Young Jaina Award* for scoring in the **top 1 percentile** overall in JEE
- Built and managed a blog on tech and marketing; reached **150K unique readers** organically.
- Secured **1st Runner-Up** in IIT Jodhpur's *Pitch It* competition for early-stage startup ideas.

REFERENCES

Dr. Devashish Gosain | Email: dgosain@cse.iitb.ac.in

Assistant Professor, Indian Institute of Technology Bombay

Dr. Ha Dao | Email: hadao@mpi-inf.mpg.de

Postdoctoral Researcher, Max Planck Institute for Informatics, INET Department

Dr. Anja Feldmann | Email: anja@mpi-inf.mpg.de

Director, Max Planck Institute for Informatics | *Group Leader*, INET Department